

FINAL PROJECT REPORT

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Team ID: Akshay-11-0g

Project Title: Food Ordering Behaviour and Consumer Trends: A Structured Analysis

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1. Introduction

1.1. Project Overview

The rapid expansion of online food delivery platforms has generated massive volumes of transactional and operational data. However, extracting actionable strategic intelligence from raw order logs remains a significant challenge for restaurant partners and delivery platform operators. This project presents an end-to-end data analytics and web integration solution titled "**Food Ordering Behaviour and Consumer Trends: A Structured Analysis.**" It processes a dataset of 50,000 order records across 27 fields to evaluate regional cuisine demand, customer demographic preferences, meal timing patterns, and logistics performance[cite: 1, 3].

1.2. Objectives

- **Demographic Segmentation:** Analyze customer ordering trends across distinct age groups and customer categories.
- **Geographic Demand Mapping:** Map cuisine popularity across major Indian metropolitan cities (Bangalore, Chandigarh, Delhi, Hyderabad, Mumbai, Pune).
- **Operational Efficiency:** Evaluate fulfillment durations to identify delivery bottlenecks and their impact on customer ratings.
- **Interactive Visualization:** Build an interactive 9-component Tableau dashboard and Tableau Story.
- **Web Integration:** Embed the published Tableau dashboard into a Python Flask web application for web browser accessibility.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement (Customer Problem Statement Template)

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A Delivery Operations Manager	Identify delivery bottlenecks and reduce customer churn	It is hard to correlate delivery delays with rating drops	Data is trapped in unorganized raw CSV files	Frustrated & stressed
PS-2	A Restaurant Partner / Strategist	Understand regional cuisine demand across cities	I don't know which items to stock in specific urban regions	I lack real-time visual insights into customer preferences	Uncertain & cautious

2.2. Project Proposal (Proposed Solution)

Project Overview

- **Objective:** Build an end-to-end analytics pipeline that converts 50,000 order logs into an interactive web dashboard[cite: 1, 3].
- **Scope:** Covers data cleaning, feature engineering, Tableau visualization, Flask web integration, and GitHub deployment.

Problem Statement

- **Description:** Delivery platforms struggle to translate high-volume transactional logs into actionable strategy.
- **Impact:** Results in unoptimized delivery routes, missed revenue during peak meal times, and lower customer ratings.

Proposed Solution

- **Approach:** Process raw CSV data, create dynamic calculated fields in Tableau, host on Tableau Public, and embed into a Flask web application[cite: 1, 2].
- **Key Features:** Top KPI Scorecard, City vs. Cuisine Heatmap, Age Group Segmentation, Delivery Duration Analysis, and Flask Web Integration[cite: 1, 2].

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware - Computing	Processor	Intel Core Ultra 5 / Intel Core i5
Hardware - Memory	RAM	16 GB DDR5 / DDR4
Hardware - Storage	Disk	SSD Storage
Software - Frameworks	Web Backend	Python Flask
Software - Libraries	Data Analytics	pandas, numpy
Software - Environment	IDE & Version Control	VS Code, Git, GitHub, Tableau Desktop / Public
Data - Source	Operational Dataset	food_ordering_behaviour_dataset.csv (50,000 rows, 27 fields)

2.3. Initial Project Planning (Product Backlog & Sprint Schedule)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date
Sprint-1	Data Pipeline	USN-1	As an analyst, I can clean missing values and fix data types in raw dataset	2	High	Akshay Chaudhari	01 Oct 2026	02 Oct 2026
Sprint-1	Feature Engineering	USN-2	As an analyst, I can create calculated fields like Age	3	High	Akshay Chaudhari	02 Oct 2026	03 Oct 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date
			Group in Tableau					
Sprint-2	Visualization	USN-3	As a manager, I can view KPI scorecards and regional heatmaps	5	High	Akshay Chaudhari	03 Oct 2026	04 Oct 2026
Sprint-2	Dashboard & Story	USN-4	As a user, I can interact with filters and view the narrative Tableau Story	3	Medium	Akshay Chaudhari	04 Oct 2026	05 Oct 2026
Sprint-3	Web Integration	USN-5	As a user, I can access the live Tableau dashboard through a Flask web app	5	High	Akshay Chaudhari	05 Oct 2026	06 Oct 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Section	Description
Project Overview	Analysis of food delivery operational logs across Indian metropolitan cities.
Data Collection Plan	Data gathered from transactional database logs exported to CSV format.

Section	Description
Raw Data Identified	Single tabular dataset containing 50,000 records across 27 fields.

Raw Data Source Details

Source Name	Description	Location / URL	Format	Size	Access Permissions
Dataset 1	Food Ordering Behaviour Dataset containing customer demographics, orders, and ratings.	Local Directory / GitHub Repository	CSV	~12 MB	Public

3.2. Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
food_ordering_behaviour_dataset.csv	Incomplete values in Delivery Time (Mins) and Customer Rating	Moderate	Imputed missing numerical fields using median values; dropped invalid records.
food_ordering_behaviour_dataset.csv	Timestamps stored as raw text strings	Low	Converted column data type to standard datetime format.
food_ordering_behaviour_dataset.csv	Unsegmented customer age numbers	Low	Created calculated dimension Age Group using conditional logic.

3.3. Data Exploration and Preprocessing

Section	Description
Data Overview	50,000 rows and 27 columns capturing transaction details, demographics, and fulfillment metrics.

Section	Description
Data Cleaning	Handled null entries, verified zero duplicate order IDs, and cleaned outlier values.
Data Transformation	Created custom calculated fields in Tableau for revenue aggregation and demographic grouping.
Data Type Conversion	Converted order times to Datetime and numerical fees/values to Float/Integer.
Column Splitting / Merging	Extracted hour of day from timestamps to categorize meal types (<i>Breakfast, Lunch, Snacks, Dinner</i>).
Data Modeling	Built internal measures and dimension hierarchies in Tableau Public.
Save Processed Data	Exported cleaned data structure and published workbook to Tableau Public.

4. Data Visualization

4.1. Framing Business Questions

1. **Question 1:** What is the overall platform volume, aggregate revenue, and average rating?
2. **Question 2:** How does cuisine preference vary geographically across major metropolitan cities?
3. **Question 3:** Which customer age groups contribute the highest order volume?
4. **Question 4:** What is the relative market share of top cuisine choices?
5. **Question 5:** How do customer ratings vary across meal timing categories?
6. **Question 6:** Which meal times generate the highest aggregate revenue?
7. **Question 7:** What is the distribution of fulfillment durations across delivery orders?
8. **Question 8:** Which cities rank highest in total order volume?

4.2. Developing Visualizations

- **Top KPI Scorecard Cards:** Displays aggregate metrics across 50,000 order records.

- **City vs. Cuisine Heatmap:** Cross-tabulation of order counts across metropolitan hubs.
- **Grouped Bar Chart:** Order distribution across Age Group and Restaurant Segments.
- **Donut Chart:** Cuisine preference percentage breakdown.
- **Pie & Stacked Bar Charts:** Customer satisfaction ratings segmented by meal type.
- **Revenue Bar Chart:** Revenue contribution by meal timing.
- **Delivery Time Histogram:** Distribution of delivery durations in minutes.
- **City Volume Bar Chart:** Ranking top cities by total order volume.

5. Dashboard

5.1. Dashboard Design File

- **Workbook Title:** food analysis (Published on Tableau Public).
- **Public Link:**
<https://public.tableau.com/views/foodanalysis/Dashboard1>

Major Dashboard Outcomes

- **Executive Overview:** Evaluates 50,000 total orders alongside aggregate platform revenue and average ratings.
- **Geographic Demand:** Fast Food and Desserts drive strong volume in Bangalore (1,446) and Mumbai (1,463), while Chinese leads in Pune (1,444) and Delhi (1,430).
- **Demographic Breakdown:** Gen Z (21 – 30) and Millennials (31 – 40) account for over 60% of total transactions.
- **Logistics Performance:** Deliveries exceeding 45 minutes lead to a sharp drop in ratings below 3.5 stars.

6. Report

6.1. Story Design File

- **Story Title:** food analysis story
- **Public Link:**
<https://public.tableau.com/views/foodanalysis/Story1>

Story Observations (City vs Cuisine Heatmap)

- **Top cuisine:** Dessert in Mumbai with 1,463 orders (highest single cell count across all city-cuisine combinations).
- **Demand over time:** Steady order distributions across major Indian cities (Bangalore, Chandigarh, Delhi, Hyderabad, Mumbai, Pune), maintaining consistent high-volume demand between 1,298 and 1,463 orders per category.
- **Value and revenue:** Fast Food and Desserts drive high transaction frequencies across metro hubs like Bangalore (1,446) and Mumbai (1,463), while Chinese cuisine shows strong regional peaks in Pune (1,444) and Delhi (1,430).
- **Recommendation:** Optimize inventory allocation and local restaurant onboarding for high-density cuisines per city—specifically expanding Dessert partnerships in Mumbai and Chinese offerings in Pune and Delhi.

7. Performance Testing

7.1. Utilization of Data Filters

- Single-select and multi-select global filters implemented for City, Cuisine Type, and Age Group across all worksheets.
- Interactive action filters configured across dashboard sheets to enable cross-filtering upon mark selection.

7.2. Number of Calculated Fields

- **3 Key Calculated Fields:** Created Age Group, Revenue Contribution, and Order Time Category.

7.3. Number of Visualizations

- **9 Distinct Worksheets:** Combined into 1 operational dashboard.

8. Conclusion / Observation

1. **Core Customer Segment:** Millennials (31 – 40) and Gen Z (21 – 30) drive over 60% of total transactions.
2. **Regional Strategy:** Platforms should optimize partner onboarding for Desserts in Mumbai and Chinese in Pune and Delhi.
3. **Logistics SLA:** Delivery operations must keep fulfillment within 45 minutes to prevent customer rating drop-offs.

9. Future Scope

1. **Predictive Analytics:** Incorporating Machine Learning models in Python to predict order volume during peak hours.
2. **Real-time Pipelines:** Connecting live database streaming (PostgreSQL / Kafka) to update Tableau visualizations dynamically.
3. **Cloud Deployment:** Deploying the Flask web application to AWS Elastic Beanstalk or Render.

10. Appendix

10.1. Source Code

app.py (Flask Backend)

Python

```
from flask import Flask, render_template
```

```
app = Flask(__name__)
```

```
@app.route('/')
```

```
def home():
```

```
    return render_template("index.html")
```

```
if __name__ == '__main__':
```

```
    app.run(debug=True)
```

templates/index.html (Frontend HTML Layout)

HTML

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
    <meta charset="UTF-8">
```

```

<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Food Ordering Analytics Dashboard</title>

<style>

    body { font-family: Arial, sans-serif; margin: 0; padding: 20px; background-color:
#f4f6f9; }

    h1 { text-align: center; color: #333; }

    .tableau-container { width: 90%; margin: 0 auto; box-shadow: 0 4px 8px rgba(0,0,0,0.1);
}

</style>

</head>

<body>

    <h1>Food Ordering Behaviour & Consumer Trends</h1>

    <div class="tableau-container">

        <iframe
src="https://public.tableau.com/views/foodanalysis/Dashboard1?:embed=yes&:showViz
Home=no" width="100%" height="850" frameborder="0"></iframe>

    </div>

</body>

</html>

```

10.2. GitHub & Project Demo Link

- **GitHub Repository Link:** <https://github.com/Akshay-11-0g/food-ordering-tableau-flask>
- **Tableau Public Demo Link:**
<https://public.tableau.com/views/foodanalysis/Dashboard1>